

Upsolving past problems

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January 16, 2026

The point of this document is for training, I will write the problems that I was unable to solve here, then come back later to solve them after having understood the idea of the solution.

CF 1060 Div 2 — C1

Problem statement: <https://codeforces.com/contest/2154/problem/C1>

HMMT Nov 2025 Teams — P5

Kelvin the frog is in the bottom-left cell of a 6×6 grid, and he wants to reach the top-right cell. He can take steps either up one cell or right one cell. However, there is a raccoon in one of the 36 cells uniformly at random, and Kelvin's path must avoid this raccoon. Compute the expected number of distinct paths Kelvin can take to reach the top-right cell.

(If the raccoon is in either the bottom-left or top-right cell, then there are 0 such paths.)

HMMT Nov 2025 Teams — P6

Let P be a point inside triangle $\triangle ABC$ such that $BP = PC$ and $\angle ABP + \angle ACP = 90^\circ$. Given that $AB = 12$, $AC = 16$, and $AP = 11$, compute the area of the concave quadrilateral $ABPC$.